

Product Specification Document - "TasteTailor"

Executive Summary

TasteTailor is an advanced, AI-powered web platform for personalized recipe management.

The application allows users to input any recipe they find online and instantly convert it to match their personal dietary preferences, lifestyle, goals, or a specific culinary style.

Furthermore, the app dynamically scales ingredients based on the number of diners and generates smart, exportable shopping lists.

To ensure users never lose a great meal, the app maintains a private, ChatGPT-style Personal Recipe History of all their tailored recipes, allowing them to bookmark specific generations into a dedicated Favorites library.

The Problem

Today, despite the abundance of culinary content online, users face significant "pain points" when trying to cook:

- Information Overload: Finding a specific recipe requires jumping between blogs and scrolling past endless text. There is no central hub tailored to the user's taste.
- Lack of Personalization: A user might find a viral recipe but cannot cook it due to dietary restrictions (vegan, celiac) or fitness goals (low-carb, high-protein).
- Fear of Unpredictable Substitutions: When home cooks substitute an ingredient (e.g., coconut oil instead of butter), they often don't understand how it will affect the texture, taste, or baking time.
- Scaling Errors: Manually calculating ingredient ratios when converting a recipe from 4 servings to 7 often leads to culinary mistakes.
- Shopping List Inefficiencies: Manually copying ingredients from a recipe to a shopping list is tedious. There is no seamless way to export these lists to third-party price-comparison bots.

- Losing Recipes: Users often lose track of recipes they loved because they are scattered across different browser bookmarks, screenshots, or forgotten in past web searches.

Target Audience

- Individuals with Dietary Restrictions: Vegans, vegetarians, people with lactose/gluten intolerances, and diabetics.
- Fitness Enthusiasts & Health-Conscious Eaters: Users tracking macros who want to convert standard recipes into high-protein or low-calorie versions.
- Culinary Enthusiasts ("Foodies"): Home cooks who love experimenting and want to give standard recipes a unique "twist" (e.g., in the style of a specific chef).
- Household Managers: People who cook for varying numbers of people and need efficient tools for meal planning and smart grocery shopping.

The Client

The primary business model at this stage is B2C (Business to Consumer).

The client is the end-user who registers to manage their daily cooking and dietary needs.

In the future, the platform can expand to a B2B2C model, allowing dietitians and personal trainers to use the tool to tailor recipes for their clients.

Business Goals

- User Retention & Engagement: Creating a seamless, highly personalized experience that turns the app into a daily/weekly habit for meal planning.
- Cost & Time Efficiency for the User: Providing exportable shopping lists that enable smart consumerism and easy price comparisons.
- Future Monetization:
 - Freemium Model: Basic recipe conversions and history storage are free, while premium features (advanced chef styles, precise nutritional macro calculations) require a subscription.
 - Affiliate Marketing: Integrating the shopping list directly with online supermarket delivery services.

Required Software Capabilities

To achieve these goals, the system requires the following technical capabilities:

1. Authentication & Profile Management: Secure login and a dedicated database table to store permanent user dietary preferences.
2. AI Integration (LLM / OpenAI API): The ability to parse raw text/URLs, apply custom prompt engineering based on user preferences, and return a structured JSON recipe.
3. "Explainable AI" Engine: The AI must generate an "Insights Box" explaining the culinary and nutritional logic behind every ingredient substitution.
4. Dynamic Frontend State Management: Real-time mathematical calculation of ingredient quantities without refreshing the page.
5. Private History Database: A robust database structure that automatically saves every AI-generated recipe to the user's private account (similar to a chat history), ensuring data is never lost.
6. Data Export System: Formatting shopping list data into structured text that can be copied to WhatsApp, Telegram bots, or price-comparison websites.

Core User Flows

Flow 1: Onboarding & Profile Setup

- The user registers securely.
- The user is directed to an initial setup screen to define their baseline profile: allergies, dietary goals (e.g., "Keto"), and general preferences.
- This data is saved and acts as the default filter for all future interactions.

Flow 2: Tailored Generation, Creator Styling & Personal History

- Input Phase: From the dashboard, the user submits either raw text or a direct URL of a recipe they wish to modify.

- Personalizing the "Twist": Users define the transformation by applying their default dietary profile (e.g., "Vegan-friendly") or selecting a specific culinary influencer to emulate.
 - Categorized Selection: A curated menu allows users to browse popular figures, such as "Gourmet Experts," "Macro-Focused Coaches," or "Plant-Based Specialists."
 - Bespoke Search: A manual entry field enables users to input the name of any creator or culinary style they prefer.
- AI Processing: The platform synthesizes the original instructions with the user's health profile and the chosen creator's persona to build a customized prompt for the AI engine.
- Output & Automatic Archiving: The resulting recipe adapts ingredients and methods to the influencer's voice. This includes an "Insights Box" detailing the rationale behind specific changes. Every tailored recipe is instantly logged in the user's Personal History for seamless future retrieval.

Flow 3: Dynamic Scaling

- On any recipe page, the default serving size is displayed (e.g., 4 servings).
- The user clicks (+/-) buttons to adjust the servings to 6.
- All ingredient quantities update proportionally and instantly on the client side.

Flow 4: Personal Favorites Library

- While viewing any recipe from their personal history, the user can click a "Heart" icon.
- This action marks the recipe as a favorite in the database.
- The user can navigate to a dedicated "Favorites" tab to view a clean gallery of their top recipes for quick, repeated access, separate from their general history.

Flow 5: Smart Shopping List & Export

- Inside a recipe page, the user clicks "Add to Shopping List."
- The system aggregates the ingredients into the user's active shopping list in the database (intelligently merging identical items, e.g., combining 2 onions and 3 onions into "5 onions").
- In the "My Shopping List" page, the user can check off items they already own.
- The user clicks "Export for Price Comparison," which generates a clean, readable text format automatically copied to the clipboard, ready to be pasted into external comparison bots or shopping platforms.